

AROGYA SAATHI

AI ASHA Worker for the World - 90+ Languages

Techfest IIT Bombay • SPARKX •

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Durjanpur, Suremanpur, Ballia, Uttar Pradesh •

AICS • IIT Patna Hybrid Program

.Tagline: AI should reach the last village, in its own language

SPEAK 

SHOW 

AI 

CARE 

THE PROBLEM

When a doctor is not immediately available

Rural families can face distance, travel, availability and language •
.barriers

People may not know whether symptoms need professional •
.attention

The gap is often: symptoms → understanding risk → reaching •
.the right facility

.Arogya Saathi targets this first-level guidance and referral gap •



THE SOLUTION

A simple, rural-first AI health companion

- Voice-first Hindi/Bhojpuri/English and ANY Languages
- .interaction
- Camera-assisted screening for possible anemia indicators and
- .throat/oral concerns
- Simple next-step guidance: monitor, visit PHC/Ayushman Arogya
- .Mandir, or seek urgent care
- .No autonomous diagnosis. No medicine prescription

SPEAK
→
UNDERSTAND
→
SCREEN
→
GUIDE
→
REFER

HOW IT WORKS

Five steps from village voice to healthcare guidance

User Presses Mic & Speaks in ANY Language .1

Bhojpuri: Hamara Pet Dard Karat Ba

Marathi: Mala Pot Dukhatay

Bengali: Amar Pet Betha Korche

French: J'ai Mal au ventre

.Whisper Large v3 Auto-Detects Language Code .2

.AI Triage Engine checks risk .3

TTS replies in SAME language + Shows PHC.4

.User receives a clear next step and referral pathway .5 •

VOICE 



STT



AI + VISION



SAFETY



PHC

AI FEATURES + SAFETY

Multimodal AI with a strong safety boundary

Possible anemia screening: visual signal only; clinical testing is •
.required for confirmation

Throat/oral screening: flags visual concerns for professional •
.evaluation

Hindi/Bhojpuri/English and ANY Language voice interaction •
.reduces typing and literacy barriers

Safety layer prevents prescription-style responses and escalates •
.warning signs

VOICE AI
VISION AI
SAFETY RULES
HUMAN CARE

UI / LIVE DEMO

Designed for simple, low-literacy interaction

HOME:  Speak |  Camera Check |  Find Care |  •
Emergency

”Example voice: “हमरा कमजोरी लागत बा। •

”.Result card: “This is a screening result, not a diagnosis •

”.Action button: “Visit PHC for further evaluation •

HOME

SPEAK 

CAMERA 

FIND CARE 

? WHY DIFFERENT

Not another generic health chatbot

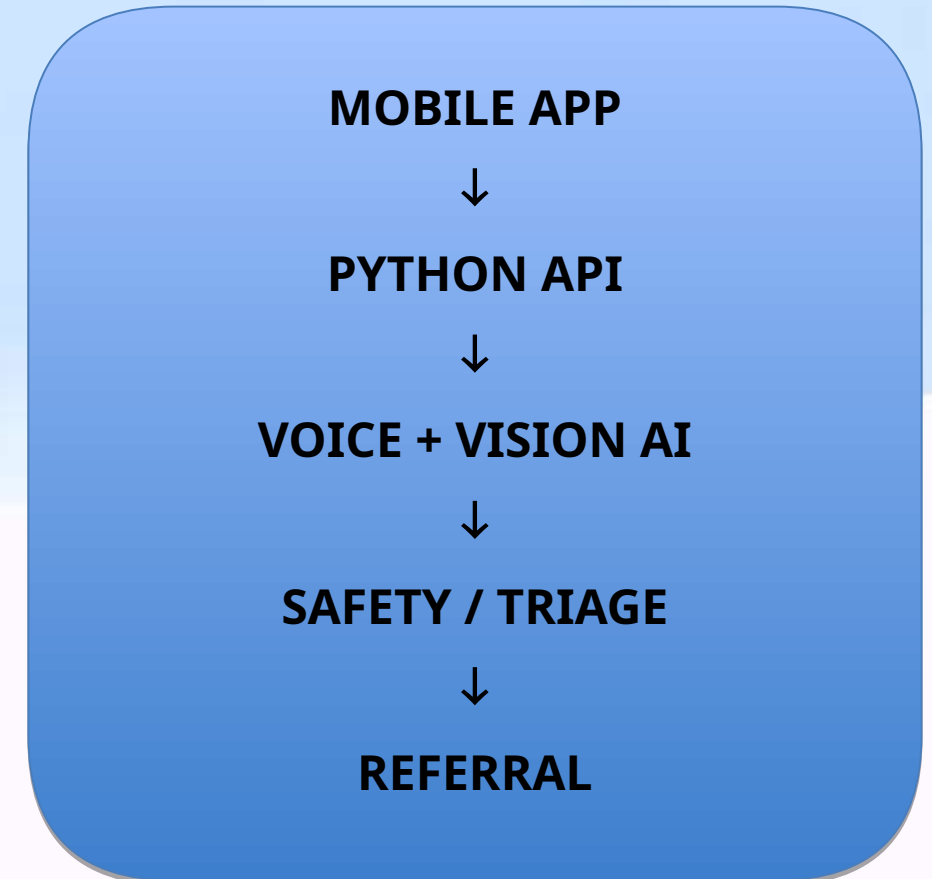
- .Rural-first: designed around village conditions
- Voice-first: Hindi/Bhojpuri/English and ANY Languages as primary interaction
- .Multimodal: voice + camera + AI + referral
- ASHA-worker support: designed as decision-support, not replacement
- Referral-focused: the goal is the safest next action, not medicine
- .prescription

GENERIC APP
TEXT / INFO
VS
AROGYA SAATHI
VOICE + VISION + REFERRAL

TECHNOLOGY

Simple prototype architecture

- .Mobile: Flutter / React Native / Android •
- .Backend: Python + FastAPI •
- AI: Whisper Large v3 (90+ World) + Bhashini (22 Indian) + Sarvan •
AI + FastAPI
- Voice: speech-to-text for Hindi/Bhojpuri/English and ANY •
- .Languages workflow
- .Data: secure, minimal-retention storage with consent •



SOCIAL IMPACT

From Durjanpur Ballia Eastern UP, Bihar, India World

- .Earlier awareness of possible health risks •
- Better navigation to PHC/Ayushman Arogya Mandir and •
- .appropriate care
- Local-language accessibility for Hindi/Bhojpuri/English ANY •
- .Languages speakers
- .Potential support for ASHA/community health workers •
- .Scalable architecture for other Indian languages and regions •

Durjanpur



BALLIA



EASTERN UP



BIHAR



INDIA

VISION

AI should reach the last village

Prototype — 2026 •

Supervised pilot + clinical/technical validation — 2027 •

Future — 90+ World Language Network + Offline Mode + Open •
API

Core principle: Arogya Saathi assists people and healthcare •
.workers; it does not replace doctors

PROTOTYPE



VALIDATE



PILOT



SCALE